((2+3i)/(4+5i)) power -2 find real and imaginary part

We need the real and imaginary parts of

$$\left(\frac{2+3i}{4+5i}\right)^{-2}$$

I. Simplify the negative power

$$\left(\frac{2+3i}{4+5i}\right)^{-2} = \left(\frac{4+5i}{2+3i}\right)^{2}$$

Step 2: Simplify $\frac{4+5i}{2+3i}$

Multiply by the conjugate of the denominator:

$$\frac{4+5i}{2+3i} \times \frac{2-3i}{2-3i}$$

Numerator:

$$(4+5i)(2-3i)$$

$$= 8 - 12i + 10i - 15i^2$$

$$= 8 - 2i + 15$$

$$= 23 - 2i$$

Denominator:

$$(2+3i)(2-3i) = 4 + 9 = 13$$

So

$$\frac{4+5i}{2+3i} = \frac{23}{13} - \frac{2}{13}i$$

Step 3: Square it

$$\left(\frac{23}{13} - \frac{2}{13}i\right)^2$$

Using $(a - bi)^2 = a^2 - b^2 - 2abi$

$$= \left(\frac{23}{13}\right)^2 - \left(\frac{2}{13}\right)^2 - 2\left(\frac{23}{13}\right)\left(\frac{2}{13}\right)i$$

$$= \frac{529}{169} - \frac{4}{169} - \frac{92}{169}i$$

$$= \frac{525}{169} - \frac{92}{169}i$$

✅ Final Answer

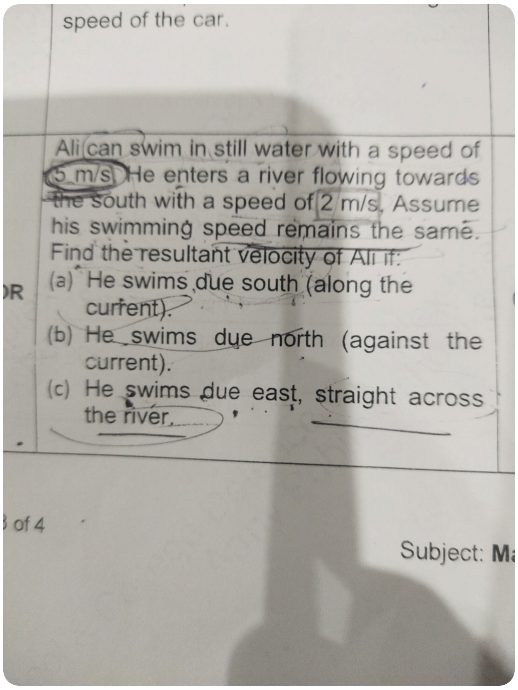
$$\left(\frac{2+3i}{4+5i}\right)^{-2} = \frac{525}{169} - \frac{92}{169}i$$

Real part

$$\boxed{\frac{525}{169}}$$

Imaginary part

$$\boxed{-\frac{92}{169}}$$



Solve it

Given:

- Speed of Ali in still water = 5 m/s
- Speed of river current = 2 m/s towards south

Resultant velocity = vector sum of swimming velocity and river velocity.

(a) He swims due south (along the current)

Both velocities are in the same direction.

$$V_R = 5 + 2 = 7 \text{ m/s}$$

✅ Resultant velocity = 7 m/s south

(b) He swims due north (against the current)

Velocities are opposite.

$$V_R = 5 - 2 = 3 \text{ m/s}$$

Direction will be north (since swimming speed is larger).

✅ Resultant velocity = 3 m/s north

(c) He swims due east (across the river)

Now the velocities are perpendicular.

East component = 5 m/s

South component = 2 m/s

Magnitude:

$$V_R = \sqrt{5^2 + 2^2}$$

$$V_R = \sqrt{25 + 4} = \sqrt{29}$$

$$V_R \approx 5.39 \text{ m/s}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{2}{5}\right)$$

$$\theta \approx 21.8^\circ$$

So the direction is 21.8° south of east.

✅ Resultant velocity = $\sqrt{29} \approx 5.39$ m/s, 21.8° south of east

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Final Answers

(a) 7 m/s south

(b) 3 m/s north

(c) 5.39 m/s, 21.8° south of east

If you want, I can also show a quick vector diagram method (1-line exam trick) to solve part (c) faster.

What is still water speed how we say east component speed is 5

Good question 👍

1. What is speed in still water?

Speed in still water means the swimmer's speed relative to the water when the water is not flowing (no current).

So here:

$$\text{Ali's speed in still water} = 5\ \text{m/s}$$

This means Ali can swim at 5 m/s in whatever direction he chooses relative to the water.

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2. Why is the east component = 5 m/s in part (c)?

In part (c) the question says:

> He swims due east

That means Ali directs his swimming toward the east.

His swimming speed relative to water = 5 m/s, so his velocity vector is:

$$5\ \text{m/s east}$$

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3. River velocity

The river flows:

$$2\ \text{m/s south}$$

So we have two perpendicular velocities:

- 5 m/s east (Ali swimming)
- 2 m/s south (river current)

---

4. Resultant velocity

These combine as vectors:

$$V = \sqrt{5^2 + 2^2}$$

✅ Key idea:
Still water speed tells us Ali's own swimming speed and direction, which is 5 m/s east when he chooses to swim east.

If you want, I can also show the vector triangle diagram (very important for exams) that makes this instantly clear. 📐

If the swimming seepd is I m/s

If Ali's swimming speed in still water = I m/s and the river flows south at 2 m/s, we again add velocities as vectors.

(a) He swims due south (along the current)

Both velocities are in the same direction.

$$V_R = 1 + 2 = 3 \text{ m/s}$$

✅ Resultant velocity = 3 m/s south

(b) He swims due north (against the current)

Opposite directions:

$$V_R = 2 - 1 = 1 \text{ m/s}$$

Since the river is stronger, the resultant is towards south.

✅ Resultant velocity = 1 m/s south

(c) He swims due east

Velocities are perpendicular:

- East = 1 m/s
- South = 2 m/s

Magnitude:

$$V = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.24 \text{ m/s}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{2}{1}\right) \approx 63.4^\circ$$

So direction is 63.4° south of east.

Final answers (when swimming speed = 1 m/s)

(a) 3 m/s south

(b) 1 m/s south

(c) 2.24 m/s, 63.4° south of east

Difference between congurent and concurrent

These two terms are used in different areas of mathematics.

1. Congruent

Congruent means exactly the same in shape and size.

- Used for figures or shapes (triangles, angles, circles, etc.).
- If two figures are congruent, they can perfectly overlap each other.

Example:

- Two triangles with the same sides and angles are congruent triangles.
- Symbol used: ≅

Example statement:

$$\triangle ABC \cong \triangle DEF$$

---

2. Concurrent

Concurrent means meeting at a single point.

- Used for lines or vectors.
- When three or more lines intersect at one point, they are called concurrent lines.

Example:

- The medians of a triangle are concurrent at the centroid.

✅ Simple difference

Term	Meaning	Used for
Congruent	Same shape and size	Shapes (triangles, angles)
Concurrent	Meet at one point	Lines

📌 Easy memory trick:

- Congruent → Same
- Concurrent → Meet together

We say that concurrent lines are not same length but parallel

No — that statement is not correct. Concurrent lines are not parallel.

Concurrent lines

Concurrent lines are three or more lines that meet at a single point.

Example:

The medians of a triangle meet at the centroid, so they are concurrent.

```
\ | /
 \|/
  •   (all lines meet at one point)
 /|\
/ | \
```

Parallel lines

Parallel lines are lines that never meet, no matter how far they extend.

```
|      |
|      |
|      |
```

Key Difference

Type	Meaning
Concurrent lines	Meet at one point
Parallel lines	Never meet

Important

Concurrent lines cannot be parallel, because parallel lines never intersect, while concurrent lines must intersect at one point.

✅ So the correct statement is:

Concurrent lines intersect at a common point, while parallel lines never intersect.